

Methods of Differentiation

Introduction

$$\begin{aligned}\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} &= \frac{dy}{dx} = f'(x) \\ &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}\end{aligned}$$

**Q: Prove that if $\rightarrow$
 $f(x) = e^x \rightarrow$ then $\rightarrow f'(x) = e^x$**

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{e^x (e^h - 1)}{h} = e^x \lim_{h \rightarrow 0} \frac{e^h - 1}{h}$$

$$f'(x) = e^x \lim_{h \rightarrow 0} \frac{e^h - 1}{h} = e^x (1) = e^x$$

Q: Prove that if $\rightarrow$

$f(x) = \sin x \rightarrow \text{then} \rightarrow f'(x) = \cos x$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{2 \cos\left(\frac{x+h+x}{2}\right) \sin\left(\frac{x+h-x}{2}\right)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\cos\left(\frac{2x+h}{2}\right) \sin\left(\frac{h}{2}\right)}{\frac{h}{2}}$$

$$f'(x) = \lim_{h \rightarrow 0} \cos\left(\frac{2x+h}{2}\right) \lim_{h \rightarrow 0} \left[\frac{\sin\left(\frac{h}{2}\right)}{\frac{h}{2}} \right]$$

$$f'(x) = \lim_{h \rightarrow 0} \cos\left(\frac{2x+h}{2}\right) (1) = \cos x$$

Standard Derivatives

$$\frac{d}{dx}(\text{constant}) = 0 \quad \frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(e^x) = e^x \quad \frac{d}{dx}(a^x) = a^x \log_e a$$

$$\frac{d}{dx}(\log_e x) = \frac{1}{x} \quad \frac{d}{dx}(\log_a x) = \frac{1}{x \log_e a}$$

$$\frac{d}{dx}(\sin x) = \cos x \quad \frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x \quad \frac{d}{dx}(\cot x) = -\csc^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

Theorem of Differentiation

1. Product Rule

$$\begin{aligned} & \frac{d}{dx} [f(x) g(x)] \\ &= f(x) \frac{d}{dx} [g(x)] + g(x) \frac{d}{dx} [f(x)] \end{aligned}$$

2. Quotient Rule

$$\begin{aligned} & \frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \\ & \frac{g(x) \frac{d}{dx} [f(x)] - f(x) \frac{d}{dx} [g(x)]}{[g(x)]^2} \end{aligned}$$

3. Chain Rule

$$\frac{d}{dx} [f(g(x))] = f'(g(x)) g'(x)$$

$$\frac{d}{dx} [\sin(x^2 + 1)] = \cos(x^2 + 1) 2x$$

$$\frac{d}{dx} [\cos(x^3)] = -\sin(x^3) 3x^2$$

Inverse Trigonometric Functions

$$\frac{d}{dx} (\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} (\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} (\tan^{-1} x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx} (\cot^{-1} x) = \frac{-1}{1 + x^2}$$

$$\frac{d}{dx} (\sec^{-1} x) = \frac{1}{|x| \sqrt{x^2 - 1}}$$

$$\frac{d}{dx} (\operatorname{cosec}^{-1} x) = \frac{-1}{|x| \sqrt{x^2 - 1}}$$

Logarithmic Differentiation

Used in functions where a variable is raised to the power of a variable.

$$f(x) = \text{variable}^{\text{variable}}$$

$$y = \sin x^{\cos x}$$

$$\log_e y = \log_e (\sin x^{\cos x})$$

$$= \cos x \log_e (\sin x)$$

$$\frac{d}{dx} (\log_e y) = \frac{d}{dx} [\cos x \log_e (\sin x)]$$

$$\frac{1}{y} \frac{dy}{dx} = \cos x \frac{d}{dx} [\log_e (\sin x)] + \log_e (\sin x) \frac{d}{dx} \cos x$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{\cos x}{\sin x} \cos x + [\log_e (\sin x)] (-\sin x)$$

$$\frac{1}{y} \frac{dy}{dx} = \cot x \cos x - \sin x \log_e (\sin x)$$

$$\frac{dy}{dx} = y [\cot x \cos x - \sin x \log_e (\sin x)]$$

$$\frac{dy}{dx} = \sin x^{\cos x} [\cot x \cos x - \sin x \log_e (\sin x)]$$

Therefore, to find the derivative of a function in which a variable is raised to a variable, apply $\log_e$ (ln) on both sides

$$y = (x - 1) (x - 2) (x + 3)$$

$$\log_e y = \log_e [(x - 1) (x - 2) (x + 3)]$$

$$\log_e y = \log_e (x - 1) + \log_e (x - 2) + \log_e (x + 3)$$

$$\begin{aligned} & \frac{d}{dx} (\log_e y) \\ &= \frac{d}{dx} [\log_e (x - 1) + \log_e (x - 2) + \log_e (x + 3)] \end{aligned}$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{x - 1} + \frac{1}{x - 2} + \frac{1}{x + 3}$$

$$\frac{dy}{dx} = y \left[\frac{1}{x-1} + \frac{1}{x-2} + \frac{1}{x+3} \right]$$

$$\frac{dy}{dx} =$$

$$(x-1)(x-2)(x+3) \left[\frac{1}{x-1} + \frac{1}{x-2} + \frac{1}{x+3} \right]$$

Differentiating a Function in Variable y With Respect To x

$$\frac{d}{dx} f(y) = \frac{d}{dx} f(y) \times \frac{dy}{dy} = \frac{d}{dy} f(y) \times \frac{dy}{dx}$$

Implicit Differentiation

An implicit function is where the x and the y terms cannot be separated.

$$x^2 + x^3y^3 - y^2 = 2y$$

$$\frac{d}{dx} (x^2 + x^3 y^3 - y^2) = \frac{d}{dx} (2y)$$

$$2x + \frac{d}{dx} (x^3 y^3) - 2y \frac{dy}{dx} = 2 \frac{dy}{dx}$$

$$2x + x^3 \frac{d}{dx} (y^3) + y^3 \frac{d}{dx} (x^3) - 2y \frac{dy}{dx} = 2 \frac{dy}{dx}$$

$$2x + x^3 (3y^2) \frac{dy}{dx} + y^3 (3x^2) - 2y \frac{dy}{dx} = 2 \frac{dy}{dx}$$

$$2x + 3x^3 y^2 \frac{dy}{dx} + 3x^2 y^3 - 2y \frac{dy}{dx} = 2 \frac{dy}{dx}$$

$$2x + 3x^2 y^3 = 2 \frac{dy}{dx} + 2y \frac{dy}{dx} - 3x^3 y^2 \frac{dy}{dx}$$

$$2x + 3x^2 y^3 = \frac{dy}{dx} (2 + 2y - 3x^3 y^2)$$

$$\frac{dy}{dx} = \frac{2x + 3x^2 y^3}{2 + 2y - 3x^3 y^2}$$

An explicit function is where the x and the y terms can be separated, and one expressed in terms of another.

$$xy = x + x^2 y$$

$$xy - x^2 y = x$$

$$y = \frac{x}{x - x^2} = \frac{1}{1 - x}$$

Concept Enhancer Question →

$$y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x \dots \infty}}}$$

$$y = \sqrt{\sin x + y}$$

$$y^2 = \sin x + y$$

$$\frac{d}{dx}(y^2) = \frac{d}{dx}(\sin x + y)$$

$$2y \frac{dy}{dx} = \cos x + \frac{dy}{dx}$$

$$2y \frac{dy}{dx} - \frac{dy}{dx} = \cos x$$

$$(2y - 1) \frac{dy}{dx} = \cos x$$

$$\frac{dy}{dx} = \frac{\cos x}{2y - 1}$$

Parametric Differentiation

$$if : x = f(t)$$

$$and : y = g(t)$$

$$then : \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

E.g., let there be 2 functions →

$$x = 4e^{-t}$$

$$y = 2e^{-t}$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\frac{d(2e^{-t})}{dt}}{\frac{d(4e^{-t})}{dt}} = \frac{2 \frac{d(e^{-t})}{dt}}{4 \frac{d(e^{-t})}{dt}}$$

$$= \frac{- (2e^{-t})}{- (4e^{-t})} = \frac{1}{2}$$

Finding the Derivative of One Function with respect to Another

E.g., find the derivative of $\rightarrow$

1. $\sin x$ with respect to $\cos x$

2. $\log_e x$ with respect to e^x

1. In the first question $\rightarrow$

$$f(x) = \sin x$$

$$g(x) = \cos x$$

$$\begin{aligned} \frac{d(f(x))}{d(g(x))} &= \frac{\frac{d(f(x))}{dx}}{\frac{d(g(x))}{dx}} = \frac{\frac{d(\sin x)}{dx}}{\frac{d(\cos x)}{dx}} \\ &= -\frac{\cos x}{\sin x} = -\cot x \end{aligned}$$

2. In the second question $\rightarrow$

$$f(x) = \log_e x$$

$$g(x) = e^x$$

$$\frac{d(f(x))}{d(g(x))} = \frac{\frac{d(f(x))}{dx}}{\frac{d(g(x))}{dx}} = \frac{\frac{d(\log_e x)}{dx}}{\frac{d(e^x)}{dx}} = \frac{1}{x e^x}$$

Higher Order Derivatives

$$\frac{dy}{dx} \rightarrow \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d^2y}{dx^2}$$

$$\frac{d^2y}{dx^2} \rightarrow \frac{d}{dx} \left(\frac{d^2y}{dx^2} \right) = \frac{d^3y}{dx^3}$$

Standard Substitutions to Make Problem Solving Easier

$$a^2 - x^2, \sqrt{a^2 - x^2} \Rightarrow x = a \sin \theta \text{ or } a \cos \theta$$

$$x^2 + a^2, \sqrt{x^2 + a^2} \Rightarrow x = a \tan \theta \text{ or } a \cot \theta$$

$$x^2 - a^2, \sqrt{x^2 - a^2} \Rightarrow x = a \sec \theta \text{ or } a \csc \theta$$

$$\frac{a+x}{a-x}, \sqrt{\frac{a+x}{a-x}} \Rightarrow x = a \cos \theta$$